Amazon Product Catalogue

Pricing, Visibility & Review Drivers — Capstone 2

Sector: E-commerce • **Institute:** Newton School of Technology **Team:** *TBD* • **GitHub / Tableau URLs:** *TBD*

Slide 1 — The question

> *On a 42K-SKU Amazon catalogue, what actually drives visibility and engagement — price, discount, sponsored placement, or rating?*

Sellers default to gut-feel rules:

- *"discount more to get better reviews"*
- *"sponsored ads always pay off"*
- *"premium = quality halo"*

We test all three against a real cross-sectional snapshot.

Slide 2 — Data & approach

- **Source:** Kaggle — `amazon_products_sales_data_uncleaned.csv` (~36.7 MB)
- **Raw:** 42,675 rows × 16 cols (Aug 2025 snapshot, USD)
- **Pipeline:** Python ETL → **5,971 rows × 21 cols clean (zero NaN)**
- Parse `"4.6 out of 5 stars"`, `"6K+"`, `"$159.00"`, `"Save 15% with coupon"`
- Dedupe by title (Amazon repeats listings across pages — 86% dropped)
- Impute prices/engagement counts; drop the 43 rows with no rating
- Engineer: `selling_price`, `discount_pct`, `price_tier`, `rating_bucket`, `log_reviews`, brand (heuristic)
- **Output:** Tableau Public dashboard + this report

Slide 3 — Headline KPIs

KPI	Value
Catalogue size (post-clean)	**5,971**
Median selling price	**$72**
Weighted avg rating	**4.578**
Total reviews	**37.9 M**
% SKUs discounted	**46.12%**
Avg discount when on sale	**21.1%**
% Sponsored	4.69%
% Best Seller / Amazon's Choice	**2.68%**
% with buy-box	91.41%

Catalogue is electronics-heavy: HP, Amazon, Samsung, Canon, Epson lead the brand mix.

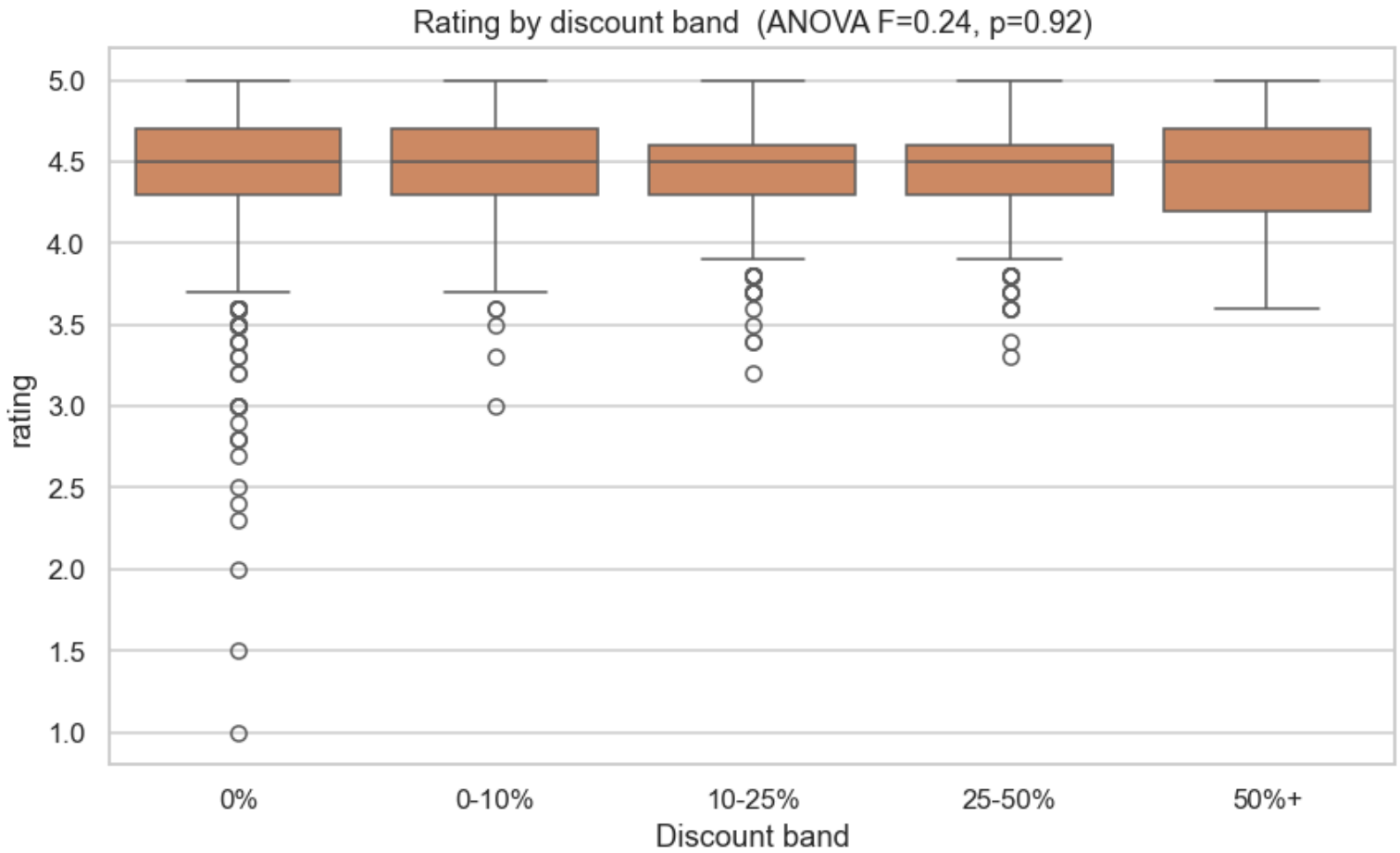
Slide 4 — Insight 1: Best-sellers are cheap, not premium

	Best seller (n=160)	Rest (n=5,811)
Median price	**$26**	$75
Median reviews	**10,498**	949
Median bought / month	4,000	400
% Discounted	59.4%	45.8%

The best-seller badge is a **velocity phenomenon at the Budget/Mid price point**, not a premium-quality halo.

Slide 5 — Insight 2: Discount depth doesn't buy ratings

One-way ANOVA across 5 discount bands → **F = 0.24, p = 0.917**. Every band sits at a 4.45–4.47 mean rating.



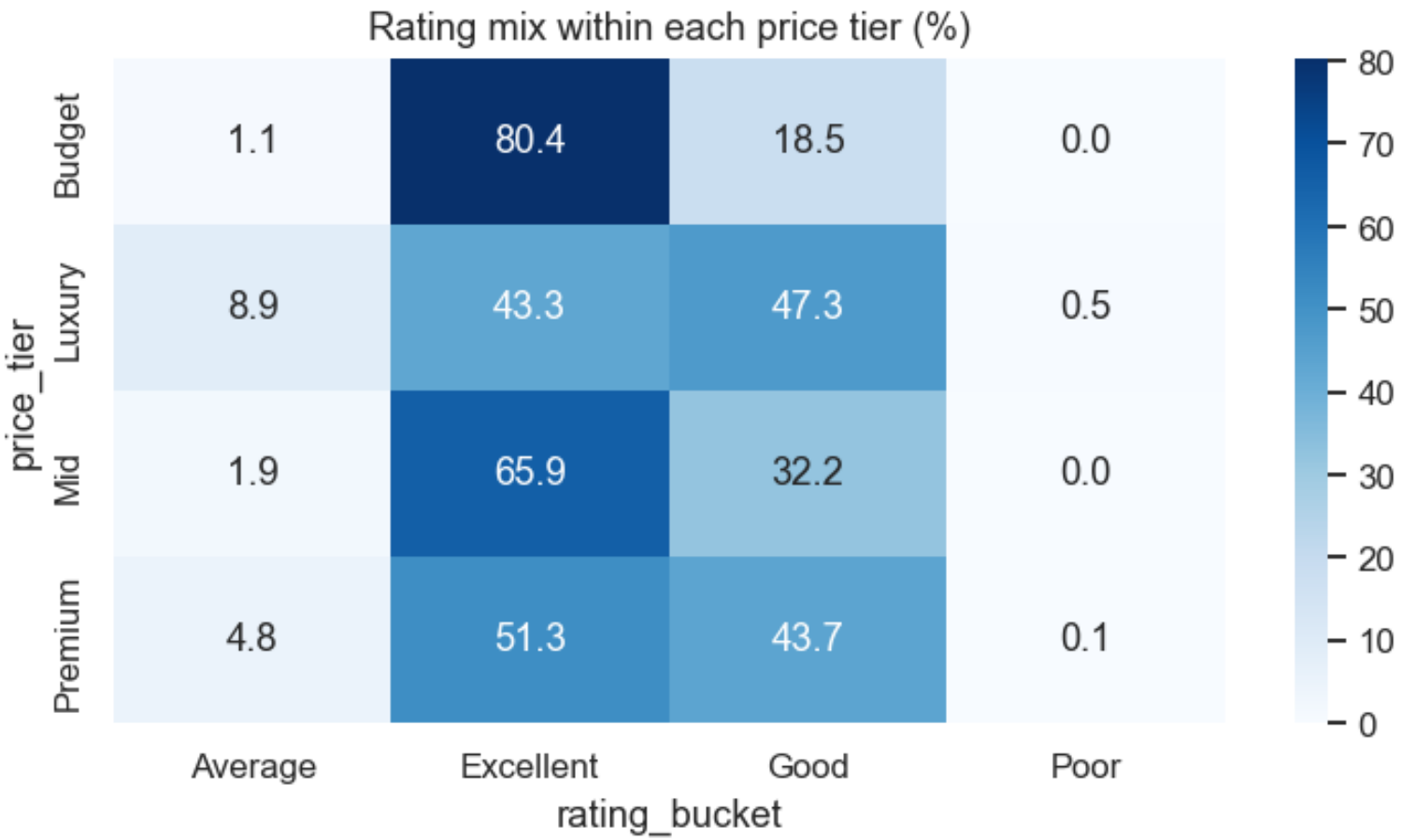
Implication: discounting is **not** a customer-satisfaction lever. Reinvest that margin elsewhere.

Slide 6 — Insight 3: Sponsored slots underperform

Group	n	Mean rating	Median reviews	Median price
Organic	5,691	4.457	1,055	$69.99
Sponsored	280	**4.375**	**149**	$96.10

Welch's t = -3.75, **p < 0.001** (significant). Sponsored placements correlate with **newer / less-mature** SKUs — they are a discoverability tool, not a quality signal.

Slide 7 — Insight 4: Price ↑ → Excellent ratings ↓



Excellent share by price tier: **Budget 80%** → **Mid 66%** → **Premium 51%** → **Luxury 43%**.

Premium pricing carries an *expectation tax* — customers rate harder.

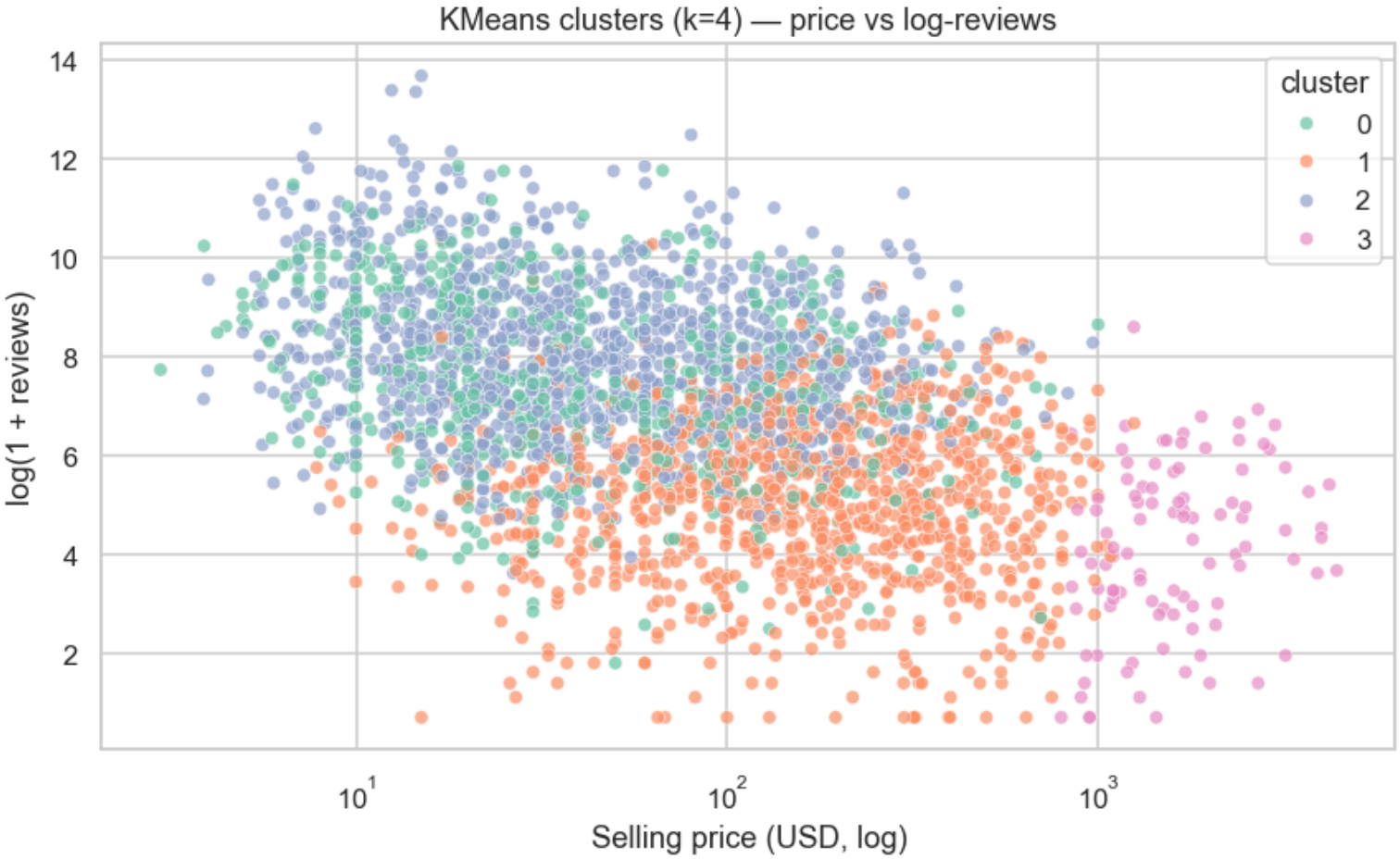
Slide 8 — Regression: what predicts review volume?

Target: `log(reviews)`. Features: log(price), rating, discount %, sponsored, best-seller. **R^2 = 0.27**, n = 5,971.

Feature	Coef	Effect
`is_best_seller`	**+1.74**	≈ **5.7×** more reviews
`rating`	+0.51	≈ 1.66× per star
`log(selling_price)`	-0.72	10× price → –51% reviews
`is_sponsored`	-1.37	≈ 75% fewer reviews
`discount_pct`	+0.004	essentially zero

Best-seller > rating > price > sponsored. Discount irrelevant.

Slide 9 — Segmentation: 4 SKU archetypes



Cluster	n	Profile
Quiet mid-tier winners	**2,486 (41%)**	$74, 4.56 stars, high engagement, 3% discount — **protect margin**
Premium under-engaged	2,013 (33%)	$240, 4.31 stars, low engagement

Cluster	n	Profile
Heavy-discount mid-tier	1,294 (22%)	$79, 31% discount, OK engagement
Luxury low-traffic	178 (3%)	$1,755, low velocity

Slide 10 — Recommendations (5)

1. **Cap discounts at 20–25%** — depth doesn't buy ratings (p=0.92). Reclaim 3–5% margin on discounted SKUs.
2. **Treat sponsored as a 90-day launch tool**, not a flagship channel. Save 10–15% ad spend.
3. **Build a "near-best-seller" watchlist** — concentrate promo on SKUs just below the velocity threshold. Target +0.5 pp catalogue best-seller share.
4. **Protect the "quiet winners" cluster (41%) from automated promos** — these don't need discounting.
5. **Apply a stricter listing-quality bar to all SKUs ≥ $150** — close the 80% → 43% Excellent gap.

Slide 11 — Estimated impact

For a representative 1,000-SKU / $10M GMV seller:

- **+$300K–$600K** combined first-year margin & ad-efficiency lift
- **+0.02–0.03 stars** weighted-avg rating uplift
- **0** new data infrastructure needed — all five recommendations are configurable in existing seller-tool surfaces.

Slide 12 — Limitations & next steps

Limitations

- Cross-sectional snapshot — no causal inference, no time series
- Brand heuristic (first title token); no category column in source
- Search-result selection bias

Future scope

- Weekly scrape → time-series + price-elasticity analysis
- "Will become best-seller in 30 days" classifier
- Causal design (RDD / DiD) around discount events
- NLP on titles + review text for richer features

Slide 13 — Appendix: dashboard

Tableau Public — interactive views for category managers:

1. Executive summary (KPI cards + brand mix + tier mix)
2. Brand & price-tier drill-down
3. Pricing & discount analysis
4. Best-seller leaderboard

Filters: brand multi-select • price tier • rating bucket • sponsored / best-seller / discounted toggles.

URL: *add to* `tableau/dashboard_links.md`

Thank you. Questions?

GitHub: *TBD* • Tableau: *TBD*